Generalized Likelihood Ratio Test for Hyperspectral Subpixel Target Detection Based on Segmented Mixing Model

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***Abstract*—For hyperspectral subpixel target detection tasks, conventional mixing model (CMM) is one of the most intensively used models. Despite the flexibility of CMM in terms of mixing coefficient, it is sometimes inappropriate to assume that all bands share the same mixing coefficient for high-dimensional spectral vectors. For the sake of finely characterizing the mixing structure of different endmembers, a segmented mixing model (SMM) for subpixel target detection is constructed. In this model, adjacent spectral bands form a segment that shares same mixing coefficient, and segments are separated from each other under some optimality. Then, a segmented-mixing-based generalized likelihood ratio test (SMGLRT) detector is developed under the framework of statistical hypothesis testing, which concentrates on solving two problems. One is to create a criterion for evaluating performance of segmentation based on block-diagonal covariance matrix, and the other is to estimate the segmented mixing coefficients and derive the GLRT statistic under the assumption of background pixels obeying Gaussian mixture distribution. Experiments on real and synthetic hyperspectral images show that the SMM-based detection outperforms than the CMM-based one, and the proposed SMGLRT detector is superior in contrast to some classical target detectors.**

***Index Terms*—Block-diagonal structure, Gaussian mixture model (GMM), generalized likelihood ratio test (GLRT), segmented mixing model (SMM), subpixel target detection.**

I. Introduction

HYPERSPECTRAL image (HSI) has received wide attention in detection and recognition tasks owing to its abundant spectral information. However, the pursuit of higher spectral resolution always comes at the expense of spatial resolution. Under such circumstance, a pixel may be composed of multiple endmembers rather than a pure one. Besides, due to the complexity of surface environment and ground objects, only two dimensions of spatial coordinates cannot completely reflect the composition at different altitudes. It indicates that the reflectance of a single pixel may be affected by the longitudinal mixing of various objects. This phenomenon is quite common in complex real-world scenes, such as urban and military fields. Whether it is horizontally mixing or vertically stacking, the phenomenon of mixed pixel has brought to the so-called subpixel target detection that confronts new challenges in practical applications.

Subpixel target can be deemed as an object that only occupies a fraction of single pixel from spatial perspective, or contributes a certain proportion to the reflectance of a pixel from spectral perspective. Detecting a subpixel target, especially imperceptible subpixel target, is always a daunting task. There are two potential issues that need to be addressed seriously. One is to detect whether a pixel under test (PUT) contains the intended target or not. Another one is to determine the proportion of the target in the PUT. To address these two issues, numerous efforts have been dedicated to designing subpixel target detection algorithms and many studies have been profoundly affected by statistical hypothesis testing theory (see [1]).

Kelly [2] developed a famous detector based on the generalized likelihood ratio test (GLRT). After that, the adaptive matched filter (AMF) [3] detector is designed for the additive model. And some extensions of AMF detector, such as elliptically-contoured finite target matched filter [4], are proposed. Besides, the one-step GLRT for replacement model is derived, which is referred to as adaptive cell under test estimator [5]. Except for GLRT-based methods, there are many other subpixel target detection approaches. One of the most widely used methods is the constrained energy minimization (CEM) [6], which detects the target by suppressing the spectral energy of background pixels in HSI. The orthogonal subspace projection [7] projects the original spectral vector into a subspace, which is orthogonal to undesired features. In [8], a detector was proposed on the basis of tree-structured encoding (TSE) which is more adaptable to different scenes. To address the issues of pure background spectra separation, insufficient target prior spectra and data noise pollution, a target prior augmentation and background suppression-based multidetector fusion (TBMF) method was proposed for detection in [9].

Apart from statistical approaches, deep learning (DL)-based methods have demonstrated remarkable performance in

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hyperspectral target detection [10], [11], [12]. Unfortunately, the lack of sufficient training sample makes DL methods difficult to capture the diversity of spectral variations [13]. To address it, the triplet hybrid attention network for target detection (THANTD) is developed based on coarse detection and radiative transfer model [14]. The triplet spectralwise transformer-based target detector (TSTTD) is proposed to construct training sample for balanced learning [15]. For the sake of maintaining the intrinsic structure of HSI while enhancing the target and background separation, the tensor adaptive reconstruction cascaded with global and local feature fusion architecture is introduced [16]. The interpretable representation network (IRN) converts physical model into a DL network and adopts a transformer module to enhance nonlinear feature representation [17]. Recently, by employing pixel attention-based linear prediction strategy and constructing band attention-based spectral abundance learning model, an attention-based sparse and collaborative spectral abundance learning subpixel target detector was presented in [18]. Moreover, a learning single spectral abundance detector is developed [19], which focuses on learning spectral abundance of the target and only requires a prior target spectrum.

To make full use of the information provided by HSI, some detectors are designed based on spatial–spectral combination. A weighted Cauchy distance graph and local adaptive collaborative representation detection method is proposed, which considers the fusion of spectral as well as spatial information [20]. The deep CEM incorporates pixel-based and cube-based deep neural network structures to further enhance the detection results [21]. A deep spatial–spectral joint-sparse prior encoding network architecture is constructed for hyperspectral target detection, which fully exploits inherent prior information and has explicit interpretability [22]. Besides, a spectral perception and spatial–temporal tensor decomposition is proposed, where both prior spectral information and spatial–temporal structural information are taken into account [23]. In order to reduce the impact of spectral variability while fully exploiting multidirection and multiscale information, a prior spectral perception and local graph fusion strategy is proposed [24].

In addition to exploring detection algorithms, a lot of work also concentrates on establishing appropriate subpixel target detection model. Among them, linear mixing model (LMM) is a physically interpretable and simple model, which has been widely employed [25]. Based on LMM, many variants have been proposed, including extended LMM [26], perturbed LMM [27], scaled and perturbed LMM [28], and generalized LMM [29]. Recently, a linear mixing and sparse representation algorithm, which incorporates subpixel target detection and unmixing processes into a united framework, is considered [30]. Integrating background modeling with discriminative subpixel target characterization, the multiple-instance subpixel adaptive cosine estimator as well as multiple-instance subpixel spectral matched filter are presented on the basis of multiple-instance learning [31]. Besides, an interaction subspace model is designed to reduce the impact of spectral variability and thereby improve robustness of subpixel target detection [32]. Under the framework of GLRT, the random projection-based subpixel target detectors are developed for three different LMMs with t-distributed background [33]. However, it is demonstrated that the subpixel target detection performance under LMM has decreased when nonlinear effects are present [34]. As for nonlinear mixing, a double dictionary-based nonlinear representation model is established based on background and target dictionaries [35]. Considering second-order scattering, bilinear mixing model (BMM) extends LMM by adding Hadamard product terms [36]. On the basis of BMM, a collaborative-guided spectral abundance learning (CGSAL) method for subpixel target detection is proposed to model nonlinear interactions among multiple materials [37]. And a BMM is utilized to synthesize nonlinear target and background spectra for training sample augmentation in transfer learning and nonlinear spectral synthesis method [38]. Further, to reflect high-order scatterings, the multilinear mixing model (MLMM) [39] as well as extended MLMM [40] are proposed. In these different patterns of mixing, some are too simple to accurately depict the mixture of background and target, while others are too sophisticated to cause overfitting. Thus, it makes great sense to construct an appropriate mixing model for subpixel target detection. And the appropriateness lies in making a tradeoff between detection performance and computation complexity.

Another challenge exists in the estimation of background covariance of high-dimensional data in HSI. For a $p \times p$ covariance matrix $\mathbf{\Sigma}$, the number of parameters to be estimated is $p(p+1)/2$ in general. It has been evidenced that the sample covariance matrix is a poor estimate when the dimension p is larger than the sample size n (see, e.g., [41]). In order to reduce the number of parameters to be estimated, a common approach is to make some structure assumptions on covariance matrix, especially block-diagonal structure which is reasonable and efficient in many practice applications (see [42]). Once the number of blocks is determined as $m (m \ll p)$ with dimension p_i of the ith block, a block diagonal structure can substantially reduce the number of parameters to $\sum_{i=1}^{m} p_i(p_i+1)/2$. Throughout this work, a block-diagonal structure of covariance matrix is taken into consideration.

This work devotes to providing a segmented-mixing-based GLRT (SMGLRT) algorithm for hyperspectral subpixel target detection, in which the segmented mixing model (SMM) plays a crucial role. The SMM is an LMM essentially as CMM, however, distinguishing from conventional LMM, its coefficients on different spectral bands are no longer completely consistent. Instead, in the SMM, the spectral bands within a segment share the same mixing coefficient, while those different-segment bands have distinct coefficients. The ingenious contributions of this work are summarized as follows.

1) The SMM is constructed to characterize mixing pattern of background and target. It can precisely reflect the degree of mixing on different spectral bands while restricting its complexity. This treatment ensures the effectiveness and robustness of the developed SMM-based detector.
2) The optimal segmentation problem is formulated as a covariance block-diagonalization task. Under the criterion of minimizing Frobenius norm, a block-diagonalization

approach of covariance matrix is developed, which behaves an advantage of efficiency for dealing with high-dimensional spectral data.

3) Two patterns of mixing are addressed and integrated in the proposed SMGLRT algorithm. The first pattern is the mixture of Gaussian distributions for characterizing more complex scene of background, and the second one aims at the mixture of endmembers for depicting actual mixed pixel phenomenon. By optimally estimating the segmented mixing coefficients and deriving the corresponding GLRT statistic for each pixel, a novel subpixel target detection for HSI is developed.

The rest of this article is organized as follows. Section II briefly reviews some related work. Section III provides the segmented-mixing-based subpixel target detection for HSI. In Section IV, the proposed SMGLRT is experimentally evaluated in a series of real HSIs as well as synthetic one. Finally, Section V concludes this article.

II. Related Work

Throughout this work, the p-dimensional spectral vectors $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n$ represent n pixels in HSI, $\text{Diag}(a_1, \ldots, a_q)$ denotes the diagonal matrix with diagonal elements $a_1, \ldots, a_q$, while $\text{BlockDiag}(\mathbf{A}_1, \ldots, \mathbf{A}_q)$ denotes the block-diagonal matrix with square matrices $\mathbf{A}_1, \ldots, \mathbf{A}_q$.

A. Gaussian Mixture Model

The probability density function of a K-component Gaussian mixture model (GMM) with each component sharing the same covariance matrix is defined as

$$p(\boldsymbol{x}; \boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \sum_{k=1}^{K} \pi_k \mathcal{N}(\boldsymbol{x}; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}) \tag{1}$$

where $\boldsymbol{\pi} = (\pi_1, \ldots, \pi_K)$ are mixing coefficients with $\pi_k \geq 0 (k = 1, \ldots, K)$ and $\sum_{k=1}^{K} \pi_k = 1$, $\boldsymbol{\mu} = (\boldsymbol{\mu}_1, \ldots, \boldsymbol{\mu}_K)$ are K mean vectors and $\boldsymbol{\Sigma}$ is the common covariance matrix of each component, and the kth Gaussian density function with mean $\boldsymbol{\mu}_k$ and covariance $\boldsymbol{\Sigma}$ is given by

$$\mathcal{N}(\boldsymbol{x}; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}) = \frac{\exp\left(-(\boldsymbol{x} - \boldsymbol{\mu}_k)^T \boldsymbol{\Sigma}^{-1} (\boldsymbol{x} - \boldsymbol{\mu}_k)/2\right)}{(2\pi)^{p/2} |\boldsymbol{\Sigma}|^{1/2}}. \tag{2}$$

Since the maximum likelihood estimation for GMM is analytically intractable, the most frequently used solution is to iteratively estimate the parameters using the expectation maximization (EM) algorithm [43]. The EM algorithm transforms the original problem into solving the maximum complete-data likelihood estimation by introducing a latent variable. Let $(\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n)$ be an independent and identically distributed sample. The parameters of GMM can be estimated by repeating the following E-step and M-step until convergence and the number of components K is determined based on the Bayesian information criterion.

1) *E-Step:*

$$\hat{\gamma}_{ik} = \frac{\hat{\pi}_k \mathcal{N}(\boldsymbol{x}_i; \hat{\boldsymbol{\mu}}_k, \widehat{\boldsymbol{\Sigma}})}{\sum_{l=1}^{K} \hat{\pi}_l \mathcal{N}(\boldsymbol{x}_i; \hat{\boldsymbol{\mu}}_l, \widehat{\boldsymbol{\Sigma}})}. \tag{3}$$

2) *M-Step:*

$$\hat{\boldsymbol{\mu}}_k = \frac{\sum_{i=1}^{n} \hat{\gamma}_{ik} \boldsymbol{x}_i}{\sum_{i=1}^{n} \hat{\gamma}_{ik}} \tag{4}$$

$$\widehat{\boldsymbol{\Sigma}} = \frac{1}{n} \sum_{i=1}^{n} \sum_{k=1}^{K} \hat{\gamma}_{ik} (\boldsymbol{x}_i - \hat{\boldsymbol{\mu}}_k)(\boldsymbol{x}_i - \hat{\boldsymbol{\mu}}_k)^T \tag{5}$$

$$\hat{\pi}_k = \frac{1}{n} \sum_{i=1}^{n} \hat{\gamma}_{ik} \tag{6}$$

where $\hat{\gamma}_{ik}$ is the estimated coefficient of $\boldsymbol{x}_i$ coming from the kth Gaussian component.

B. Subpixel Target Detection Models

Detecting subpixel targets can be transformed into solving a binary-composite hypothesis testing problem. A PUT is a pure background pixel under the null hypothesis, while it is a mixed pixel composed of background and target endmembers under the alternative hypothesis. According to different mixing patterns, it can be divided into linear and nonlinear mixing subpixel target, respectively. On the basis of LMM, three classical subpixel target detection models have been developed [44], [45], [46]. Let $\boldsymbol{t}$ and $\boldsymbol{b}$ be the target and background spectral signature, respectively. And the three models are given as follows.

1) *Additive Model:*

$$H_0 : \boldsymbol{x} = \boldsymbol{b} \quad \text{versus} \quad H_1 : \boldsymbol{x} = \alpha \boldsymbol{t} + \boldsymbol{b} \tag{7}$$

where $\alpha > 0$ is the mixing coefficient.

2) *Replacement Model:*

$$H_0 : \boldsymbol{x} = \boldsymbol{b} \quad \text{versus} \quad H_1 : \boldsymbol{x} = \alpha \boldsymbol{t} + (1 - \alpha) \boldsymbol{b} \tag{8}$$

where mixing coefficient $\alpha \in (0, 1)$.

3) *Mixing Model:*

$$H_0 : \boldsymbol{x} = \boldsymbol{b} \quad \text{versus} \quad H_1 : \boldsymbol{x} = \alpha \boldsymbol{t} + \beta \boldsymbol{b} \tag{9}$$

where $\alpha > 0$ and $\beta \geq 0$ are the mixing coefficients of background and target signatures. It is noteworthy that model (9) degenerates into model (7) and model (8) when $\alpha > 0, \beta = 1$ and $\alpha > 0, \beta = 1 - \alpha$, respectively.

III. Segmented Mixing Model for Subpixel Target Detection

A. Segmented Mixing Model

In the traditional mixing model (9), each spectral band in a pixel shares the same mixing coefficient [see Fig. 1(a)]. It is obviously not suitable for more general cases. For the sake of depicting more complicated mixing scenes, it is more appropriate to assume that all bands are assigned with different mixing coefficients, and formulate the target detection as the following hypothesis testing problem:

$$H_0 : \boldsymbol{x} = \boldsymbol{b} \quad \text{versus} \quad H_1 : \boldsymbol{x} = \mathbf{A} \boldsymbol{t} + \mathbf{B} \boldsymbol{b} \tag{10}$$

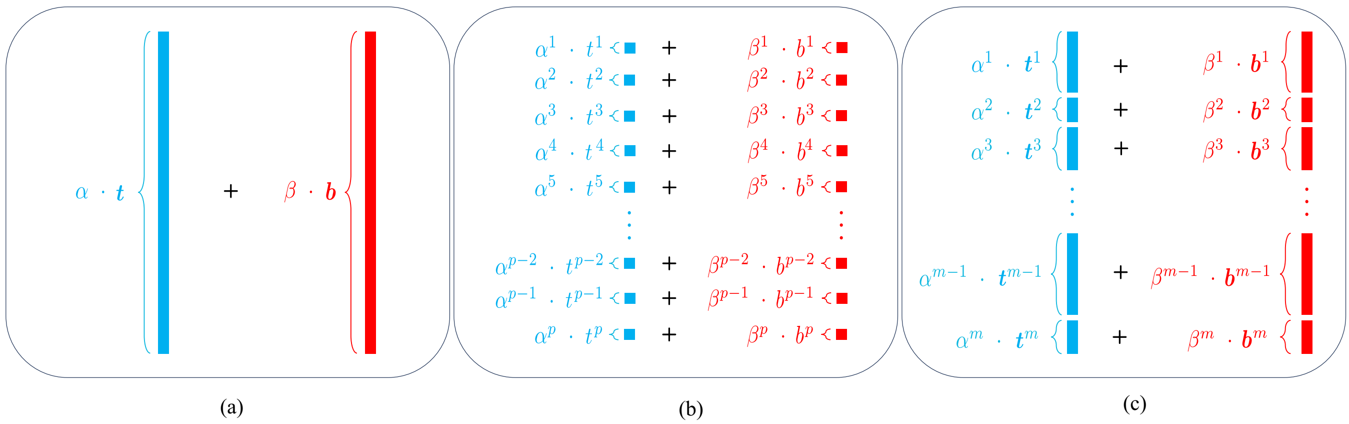


Fig. 1. Three mixing patterns of pixels. (a) Traditional mixing. (b) Fully segmented mixing. (c) Partially segmented mixing.

where $\mathbf{A} = \text{Diag}(\alpha^1, \ldots, \alpha^p), \mathbf{B} = \text{Diag}(\beta^1, \ldots, \beta^p)$ are two diagonal matrices as mixing coefficients, and α^j and β^j represent the coefficients of background and target vectors on the jth spectral band ($j = 1, \ldots, p$). As a comparison, in (9), the mixing coefficients α and β are only both scalars.

Model (10) can be employed for fully segmented mixing case (i.e., linear mixtures on all spectral bands), but still confronts the problem of too many parameters. As shown in Fig. 1(b), there are a total of $2p$ parameters to be estimated. In order to improve the precision of subpixel detection model while restricting its complexity, one feasible measure is to let variables with strong correlation share same mixing coefficient, while those irrelevant variables possess individual coefficients. Under this strategy, we introduce a partially SMM [see Fig. 1(c)] and consider the following target detection problem with block-diagonal structure:

$$H_0 : \boldsymbol{x} = \boldsymbol{b} \quad \text{versus} \quad H_1 : \boldsymbol{x} = \widetilde{\mathbf{A}}\boldsymbol{t} + \widetilde{\mathbf{B}}\boldsymbol{b} \tag{11}$$

where $\widetilde{\mathbf{A}} = \text{BlockDiag}(\alpha^1\mathbf{I}_{p_1}, \ldots, \alpha^m\mathbf{I}_{p_m})$ and $\widetilde{\mathbf{B}} = \text{BlockDiag}(\beta^1\mathbf{I}_{p_1}, \ldots, \beta^m\mathbf{I}_{p_m})$ are two block-diagonal matrices as mixing coefficients, m is the number of blocks ($m \ll p$), $\mathbf{I}_{p_j}$ is the $p_j \times p_j$ identity matrix, p_j are the total number of variables in the jth band with $\sum_{j=1}^{m} p_j = p$, and α^j and β^j are mixing coefficients of the jth spectral band, $j = 1, \ldots, m$.

Fig. 1(a)–(c) illustrate the three mixing models (9)–(11), respectively. In general, model (9) is too simple to depict complex scenes, while model (10) is too elaborate and prone to overfitting. By contrast, the proposed model (11) falls somewhere in between. It can commendably capture the correlation among bands and is just right for characterizing the degree of mixing. Therefore, hypothesis test (11) is adopted as the subpixel target detection model in this work. The background spectral vector $\boldsymbol{b}$ obeys a GMM $G_p(\{\pi_k, \boldsymbol{\mu}_k, \boldsymbol{\Sigma}\}_{k=1}^K)$, which is composed of K components with each component sharing the same covariance matrix $\boldsymbol{\Sigma}$. After determining the appropriate model, there are still two pivotal issues to be addressed. One is to obtain the optimal segmentation, and the other is to derive the test statistic.

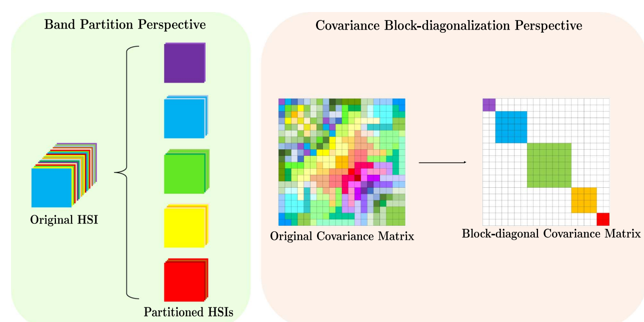


Fig. 2. Band partition and covariance block-diagonalization.

B. Optimal Segmentation

One of the important tasks in solving hypothesis test problem (11) is to find an optimal way to segment the spectrum. Intuitively, spectral segmentation is just like band partition, where an original HSI is partitioned into several multispectral images on spectral dimension. This process can also be explained from the perspective of covariance block-diagonalization (see Fig. 2), where similar bands are divided into same block.

In this work, the optimal segmentation is designed on the basis of the optimal covariance block-diagonalization. To find the optimal segmentation, the first step is to estimate the number of segments. Determining the number of clusters or blocks has always been a challenging problem. Unfortunately, this interesting topic exceeds the scope of this article. Adopting the strategy in [47], we determine the number of blocks m by implementing graph spectral clustering on bands and maximizing the relative eigengap, i.e.,

$$\hat{m} = \underset{m \in \{2, \ldots, p-1\}}{\arg\max} \frac{\lambda_{(m+1)} - \lambda_{(m-1)}}{\lambda_{(m)}} \tag{12}$$

where $\lambda_{(m)}$ denotes the mth smallest eigenvalue of the Laplacian matrix of the graph introduced in [47].

Let $\mathcal{B}$ denote the set of all partitioning ways. For any partition $B \in \mathcal{B}$, the covariance matrix in block structure is denoted as $(\boldsymbol{\Sigma}_{ij})_{i,j=1,\ldots,m}$, where $\boldsymbol{\Sigma}_{ij}$ is the cross-covariance of the ith and jth partitioned spectral bands, and the corresponding block-diagonal covariance matrix is denoted as

$$\boldsymbol{\Sigma}_B = \mathrm{BlockDiag}(\boldsymbol{\Sigma}_{11}, \ldots, \boldsymbol{\Sigma}_{mm}). \tag{13}$$

It is worth pointing out that the cardinality $|\mathcal{B}|$ will be extremely large when the number of blocks m is large. Therefore, it is always necessary to find an optimal segmentation within a smaller subset $\widetilde{\mathcal{B}}$ [48]. In this work, we randomly initialize 100 times and utilize spectral clustering to obtain 100 partitioning ways as elements in the subset $\widetilde{\mathcal{B}}$. Then, the best block-diagonal covariance matrix can be obtained by minimizing the Frobenius norm of difference between the partitioned covariance matrix and its block-diagonal form, i.e.,

$$\begin{aligned}\underset{\boldsymbol{\Sigma}_B \in \mathcal{S}_{\widetilde{\mathcal{B}}}}{\arg\min} \|\boldsymbol{\Sigma}_B - \boldsymbol{\Sigma}\|_{\mathrm{F}}^2 &= \underset{\boldsymbol{\Sigma}_B \in \mathcal{S}_{\widetilde{\mathcal{B}}}}{\arg\min} \sum_{j=2}^{m}\sum_{i=1}^{j-1} \mathrm{tr}(\boldsymbol{\Sigma}_{ji}\boldsymbol{\Sigma}_{ij}) \\ &= \underset{\boldsymbol{\Sigma}_B \in \mathcal{S}_{\widetilde{\mathcal{B}}}}{\arg\min} \left(\mathrm{tr}\boldsymbol{\Sigma}^2 - \sum_{j=1}^{m}\mathrm{tr}\boldsymbol{\Sigma}_{jj}^2\right) \end{aligned} \tag{14}$$

where $\mathcal{S}_{\widetilde{\mathcal{B}}} = \{\boldsymbol{\Sigma}_B\}_{B\in\widetilde{\mathcal{B}}}$ is the set of block-diagonal covariance matrices corresponding to candidate partitions in $\widetilde{\mathcal{B}}$.

Note that the objective function in (14) contains two unknown terms $\mathrm{tr}\boldsymbol{\Sigma}^2$ and $\mathrm{tr}\boldsymbol{\Sigma}_{jj}^2 (j = 1, \ldots, m)$, an appropriate estimate of $\mathrm{tr}\boldsymbol{\Sigma}^2 - \sum_{j=1}^{m}\mathrm{tr}\boldsymbol{\Sigma}_{jj}^2$ under the assumption of GMM needs to be addressed. Inspired by the results of the block-diagonal covariance structure for high-dimensional data in [49], we adopt an unbiased estimate of the objective function of (14) as

$$\begin{aligned} T_B = &\frac{n^2}{(n-K-1)(n-K+2)}\left[\left(\mathrm{tr}\mathbf{S}^2 - \frac{1}{n-K}(\mathrm{tr}\mathbf{S})^2\right)\right. \\ &\left. - \sum_{j=1}^{m}\left(\mathrm{tr}\mathbf{S}_{jj}^2 - \frac{1}{n-K}(\mathrm{tr}\mathbf{S}_{jj})^2\right)\right] \end{aligned} \tag{15}$$

where $\mathbf{S} = \widehat{\boldsymbol{\Sigma}}$ is the sample covariance matrix, $\mathbf{S}_{jj}$ denotes the jth $p_j \times p_j$ submatrix on the main diagonal block of $\mathbf{S}$ $(j = 1, \ldots, m)$, and n and K are the sample size and the number of components in GMM, respectively. And then, the optimal segmentation is formulated as the following optimization problem:

$$\boldsymbol{\Sigma}_B^* = \underset{\boldsymbol{\Sigma}_B \in \mathcal{S}_{\mathcal{B}}}{\arg\min}\, T_B. \tag{16}$$

C. SMM-Based GLRT

After obtaining the optimal segmentation and corresponding block-diagonal covariance matrix, we can now derive the test statistic of hypotheses (11) under the framework of GLRT.

Under the assumption of the covariance matrix $\boldsymbol{\Sigma}$ being block-diagonal, the variables from different blocks are independent of each other. So, the density function of the kth Gaussian distribution $(k = 1, \ldots, K)$ can be written as

$$N_p(\cdot; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}) = \prod_{j=1}^{m} N_{p_j}(\cdot; \boldsymbol{\mu}_k^j, \boldsymbol{\Sigma}_{jj}) \tag{17}$$

where $\boldsymbol{\mu}_k^j$ is a p_j-dimensional spectral vector corresponding to the jth block in $\boldsymbol{\mu}_k$, $j = 1, \ldots, m$.

For a PUT $\boldsymbol{x}_i$, its logarithmic probability densities under H_0 and H_1, respectively, are

$$L_0 = \ln\left(\sum_{k=1}^{K}\pi_k\prod_{j=1}^{m}N_{p_j}\left(\boldsymbol{x}_i^j; \boldsymbol{\mu}_k^j, \boldsymbol{\Sigma}_{jj}\right)\right) \tag{18}$$

$$L_1 = \ln\left(\sum_{k=1}^{K}\pi_k\prod_{j=1}^{m}N_{p_j}\left(\boldsymbol{x}_i^j; \alpha_i^j\boldsymbol{t}^j + \beta_i^j\boldsymbol{\mu}_k^j, \left(\beta_i^j\right)^2\boldsymbol{\Sigma}_{jj}\right)\right) \tag{19}$$

where, $\boldsymbol{x}_i^j$ and $\boldsymbol{t}^j$ are both p_j-dimensional spectral vectors corresponding to the jth block in $\boldsymbol{x}_i$ and $\boldsymbol{t}$, respectively, and α_i^j and β_i^j are the mixing coefficients of the jth block with respect to the PUT $\boldsymbol{x}_i$.

Similar to the estimation process of GMM, it is also impossible to directly estimate the mixing coefficients. An alternative solution is to refer to the EM algorithm to find the tightest lower bounds of L_0 and L_1, denoted as $\widetilde{L}_0$ and $\widetilde{L}_1$, respectively. Bringing in the posterior probability γ_{ik} obtained by (3), omitting the constant term and multiplying a constant, we have

$$\widetilde{L}_0 = -\sum_{j=1}^{m}\left(\sum_{k=1}^{K}\gamma_{ik}\left(\widetilde{\boldsymbol{x}}_i^j - \widetilde{\boldsymbol{\mu}}_k^j\right)^T\left(\widetilde{\boldsymbol{x}}_i^j - \widetilde{\boldsymbol{\mu}}_k^j\right)\right) \tag{20}$$

$$\begin{aligned}\widetilde{L}_1 = -\sum_{j=1}^{m}&\left(\sum_{k=1}^{K}\frac{\gamma_{ik}\left(\boldsymbol{\nu}_{ik}^j - \alpha_i^j\widetilde{\boldsymbol{t}}^j\right)^T\left(\boldsymbol{\nu}_{ik}^j - \alpha_i^j\widetilde{\boldsymbol{t}}^j\right)}{\left(\beta_i^j\right)^2}\right. \\ &\left. + p_j\ln\left(\beta_i^j\right)^2\right)\end{aligned} \tag{21}$$

where $\widetilde{\boldsymbol{x}}_i^j = \boldsymbol{\Sigma}_{jj}^{-1/2}\boldsymbol{x}_i^j$, $\widetilde{\boldsymbol{t}}^j = \boldsymbol{\Sigma}_{jj}^{-1/2}\boldsymbol{t}^j$, $\widetilde{\boldsymbol{\mu}}_k^j = \boldsymbol{\Sigma}_{jj}^{-1/2}\boldsymbol{\mu}_k^j$, and $\boldsymbol{\nu}_{ik}^j = \widetilde{\boldsymbol{x}}_i^j - \beta_i^j\widetilde{\boldsymbol{\mu}}_k^j$.

Letting the partial derivative of $\widetilde{L}_1$ with respect to α_i^j equal to 0, we obtain the maximum likelihood estimate of α_i^j as

$$\widehat{\alpha}_i^j = \frac{\sum_{k=1}^{K}\gamma_{ik}\left(\boldsymbol{\nu}_{ik}^j\right)^T\widetilde{\boldsymbol{t}}^j}{(\widetilde{\boldsymbol{t}}^j)^T\widetilde{\boldsymbol{t}}^j}. \tag{22}$$

And then, substituting (22) into (21), we have

$$\widetilde{L}_1 = -\sum_{j=1}^{m}\left(p_j\ln\left(\beta_i^j\right)^2 + \sum_{k=1}^{K}\frac{\gamma_{ik}\left(\boldsymbol{\nu}_{ik}^j\right)^T\mathbf{P}_{jj}^{\perp}\boldsymbol{\nu}_{ik}^j}{\left(\beta_i^j\right)^2}\right) \tag{23}$$

where $\mathbf{P}_{jj}^{\perp} = \mathbf{I}_{p_j} - \widetilde{\boldsymbol{t}}^j((\widetilde{\boldsymbol{t}}^j)^T\widetilde{\boldsymbol{t}}^j)^{-1}(\widetilde{\boldsymbol{t}}^j)^T$.

Let the partial derivative of the above equation with respect to β_i^j equal to 0, we immediately have

$$p_j(\beta_i^j)^2 + \sum_{k=1}^{K}\gamma_{ik}(\widetilde{\boldsymbol{\mu}}_k^j)^T\mathbf{P}_{jj}^{\perp}\widetilde{\boldsymbol{x}}_i^j\beta_i^j - \sum_{k=1}^{K}\gamma_{ik}(\widetilde{\boldsymbol{x}}_i^j)^T\mathbf{P}_{jj}^{\perp}\widetilde{\boldsymbol{x}}_i^j = 0 \tag{24}$$

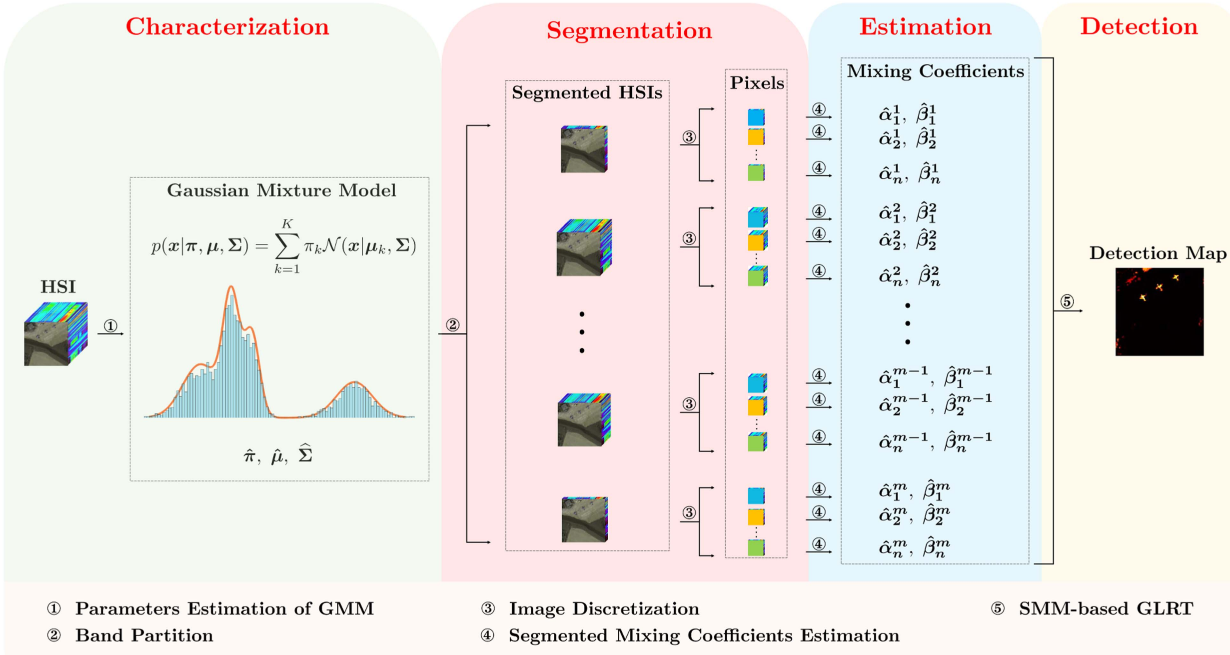


Fig. 3. Flowchart of the proposed SMGLRT detector.

and then obtain its nonnegative solution as

$$\beta_A = \frac{-\beta_B + \sqrt{\beta_B^2 + 4p_j\beta_C}}{2p_j} \tag{25}$$

where

$$\beta_B = \sum_{k=1}^{K} \gamma_{ik} \left(\tilde{\boldsymbol{\mu}}_k^j\right)^T \mathbf{P}_{jj}^{\perp} \tilde{\boldsymbol{x}}_i^j,\ \beta_C = \sum_{k=1}^{K} \gamma_{ik} \left(\tilde{\boldsymbol{x}}_i^j\right)^T \mathbf{P}_{jj}^{\perp} \tilde{\boldsymbol{x}}_i^j. \tag{26}$$

Moreover, from the constraint on mixing coefficient $\alpha_i^j \geq 0$, β_i^j should satisfy

$$\beta_i^j \leq \beta_D \stackrel{\text{def}}{=} \frac{\sum_{k=1}^{K} \gamma_{ik} \left(\tilde{\boldsymbol{x}}_i^j\right)^T \tilde{\boldsymbol{t}}^j}{\sum_{k=1}^{K} \gamma_{ik} \left(\tilde{\boldsymbol{\mu}}_k^j\right)^T \tilde{\boldsymbol{t}}^j}. \tag{27}$$

In result, the estimate of β_i^j is given by

$$\widehat{\beta}_i^j = \min\left(\beta_A, \beta_D\right). \tag{28}$$

By inputting the estimated parameters, we provide a GLRT statistic for hypotheses (11) at the PUT $\boldsymbol{x} = \boldsymbol{x}_i$ as

$$\Lambda(\boldsymbol{x}_i) = -\sum_{j=1}^{m} \left(\sum_{k=1}^{K} \frac{\gamma_{ik}}{\left(\widehat{\beta}_i^j\right)^2} \left(\widehat{\boldsymbol{\nu}}_{ik}^j\right)^T \mathbf{P}_{jj}^{\perp} \widehat{\boldsymbol{\nu}}_{ik}^j + p_j \ln\left(\widehat{\beta}_i^j\right)^2 - \sum_{k=1}^{K} \gamma_{ik} \left(\tilde{\boldsymbol{x}}_i^j - \tilde{\boldsymbol{\mu}}_k^j\right)^T \left(\tilde{\boldsymbol{x}}_i^j - \tilde{\boldsymbol{\mu}}_k^j\right) \right) \tag{29}$$

where $\widehat{\boldsymbol{\nu}}_{ik}^j = \tilde{\boldsymbol{x}}_i^j - \widehat{\beta}_i^j \tilde{\boldsymbol{\mu}}_k^j$.

Algorithm 1: SMGLRT.

Input: The HSI pixels $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n$, the spectral signature of target $\boldsymbol{t}$.
Output: The detection outputs $\Lambda(\boldsymbol{x}_1), \ldots, \Lambda(\boldsymbol{x}_n)$.
Begin:
Choose the number of components K on the foundation of Bayesian information criterion;
repeat
  Update coefficients $\hat{\gamma}$ using Eq. (3);
  Update mean vectors $\hat{\boldsymbol{\mu}}$ using Eq. (4);
  Update covariance matrix $\widehat{\boldsymbol{\Sigma}}$ using Eq. (5);
  Update proportions of mixture $\hat{\pi}$ using Eq. (6);
until Convergence;
Determine the number of blocks m by adopting spectral clustering and relative eigengap maximization criterion;
Construct a set of segmentations $\mathcal{B}$ through random initialization and spectral clustering;
Obtain the optimal segmentation according to Eq. (16);
for $i = 1 \to n$ **do**
  for $j = 1 \to m$ **do**
    Estimate $\widehat{\beta}_i^j$ using Eq. (28);
    Estimate $\widehat{\alpha}_i^j$ using Eq. (22);
  end for
  Compute the detection output $\Lambda(\boldsymbol{x}_i)$ using Eq. (29);
end for
End

Repeatedly applying (29) for all $i = 1, \ldots, n$, the hyperspectral subpixel target detection can be performed on all pixels. The detailed detection procedure of the proposed SMGLRT for an HSI is summarized in Algorithm 1.

The whole process of SMM-based hyperspectral subpixel target detection is also illustrated in Fig. 3. It clearly indicates that the SMGLRT-based detection can be roughly divided into four parts: 1) characterization; 2) segmentation; 3) estimation; and 4) detection.

In the characterization part, the GMM is utilized to characterize the background distribution, and the computational complexity of $\mathcal{O}(npKT)$ is needed for estimating parameters, where n, p, K, and T refer to the numbers of pixels, spectral bands, Gaussian components, and iterations, respectively. Then, partitioning bands into several segments requires $\mathcal{O}(p^2m)$ operations, where m is the number of segments. As for the third part, the estimation of segmented mixing coefficients needs $\mathcal{O}(nK\sum_{j=1}^{m}p_j^2)$ operations. The complexity of detection part is also $\mathcal{O}(nK\sum_{j=1}^{m}p_j^2)$. Thus, the total computational complexity of the proposed SMGLRT method is $\mathcal{O}(npKT + p^2m + nK\sum_{j=1}^{m}p_j^2 + nK\sum_{j=1}^{m}p_j^2)$. In particular, the complexity can be approximately scaled as $\mathcal{O}(nK\sum_{j=1}^{m}p_j^2)$ when $pT \approx \sum_{j=1}^{m}p_j^2$.

It is also noteworthy that the approximate computational complexity of the SMM-based detection is $\mathcal{O}(nK\sum_{j=1}^{m}p_j^2)$, while that of the CMM-based one is $\mathcal{O}(nKp^2)(\sum_{j=1}^{m}p_j^2 \ll p^2)$. Thus, adopting SMM is theoretically more efficient.

IV. Experiments and Analysis

In this section, all experiments are tested on a platform equipped with GeForce RTX 4090D GPU and Intel Xeon Platinum 8481 C CPU. To evaluate the effectiveness of the proposed detector, several hyperspectral target detection scenarios are considered.

A. Datasets Description and Experimental Setting

1) *SDA*[1]: This HSI was acquired by the airborne visible/infrared imaging spectrometer (AVIRIS) sensor. The scene of HSI is the San Diego airport. This article selects a region of interest with size of $100 \times 100 \times 189$. Three aircraft occupying 57 pixels in SDA are regarded as targets.
2) *Cri*[2]: The Cri HSI was collected by the Nuance Cri hyperspectral sensor [50]. It covers an area of 400×400 pixels and has 46 spectral bands. In this scene, ten rocks with different sizes are considered as the targets, which contains 2216 pixels.
3) *WTC*[3]: Another real HSI was also captured by the AVIRIS sensor at the world trade center on September 16, 2001. The spatial size of the image is 200×200 and the number of spectral channels is 224. In this HSI, fire sources consisting of 83 pixels are taken as targets to be detected [51].
4) *Hyperion*[4]: The Hyperion HSI covers a rural area in Indiana, USA [52]. This image contains 150×150 pixels with 155 spectral channels. The roof and silo are considered as targets, which occupies 17 pixels.
5) *Pavia*[4]: This HSI was obtained by the reflective optics system imaging spectrometer (ROSIS) sensor at the city center of Pavia, Italy [53]. The size of this experimental HSI is $100 \times 100 \times 102$. The vehicles on the bridge are treated as targets, with a total of 71 target pixels.
6) *SYN:* This HSI is a synthetic image with size of $280 \times 280 \times 126$, which is generated according to the linear mixing method in [47]. 25 targets with different sizes, shapes as well as intensities are implanted in a pure background image.

To assess the performance of proposed detector, six target detectors including AMF [3], CEM [6], IRN [17], TSTTD [15], TSE [8] and CGSAL [37] are adopted for comparison. The parameters in these detectors are set according to their original articles and tuned to achieve their best detection performance.

On the basis of classical receiver operating characteristic (ROC) curve, the 3-D ROC curve is introduced [54], which can further evaluate the target detectability (TD), background suppressibility (BS), overall detection (OD), OD probability (ODP), and signal-to-noise probability ratio (SNPR) of a detector. This work adopts 3-D ROC curve and nine area under curves (AUCs), which are $\text{AUC}_{(\text{TPR, FPR})}$, $\text{AUC}_{(\tau,\text{TPR})}$, $\text{AUC}_{(\tau,\text{FPR})}$, AUC_{TD}, AUC_{BS}, $\text{AUC}_{\text{TD-BS}}$, AUC_{OD}, AUC_{ODP}, and AUC_{SNPR}, to quantitatively evaluate the effectiveness of detectors. Three measures TPR, FPR, and τ stand for true positive rate, false positive rate, and threshold, respectively. For the calculation of $\text{AUC}_{(\text{TPR, FPR})}$, $\text{AUC}_{(\tau,\text{TPR})}$, and $\text{AUC}_{(\tau,\text{FPR})}$ values, see [54]. Then, the calculation formulas of AUC_{TD}, AUC_{BS}, $\text{AUC}_{\text{TD-BS}}$, AUC_{OD}, AUC_{ODP}, and AUC_{SNPR} are given as

$$\text{AUC}_{\text{TD}} = \text{AUC}_{(\text{TPR, FPR})} + \text{AUC}_{(\tau,\text{TPR})} \tag{30}$$

$$\text{AUC}_{\text{BS}} = \text{AUC}_{(\text{TPR, FPR})} - \text{AUC}_{(\tau,\text{FPR})} \tag{31}$$

$$\text{AUC}_{\text{TD-BS}} = \text{AUC}_{(\tau,\text{TPR})} - \text{AUC}_{(\tau,\text{FPR})} \tag{32}$$

$$\text{AUC}_{\text{ODP}} = \text{AUC}_{(\tau,\text{TPR})} + (1 - \text{AUC}_{(\tau,\text{FPR})}) \tag{33}$$

$$\text{AUC}_{\text{OD}} = \text{AUC}_{(\text{TPR, FPR})} + \text{AUC}_{(\tau,\text{TPR})} - \text{AUC}_{(\tau,\text{FPR})} \tag{34}$$

$$\text{AUC}_{\text{SNPR}} = \frac{\text{AUC}_{(\tau,\text{TPR})}}{\text{AUC}_{(\tau,\text{FPR})}}. \tag{35}$$

B. Comparison of the CMM and SMM

One of the main contributions in this work is proposing an SMM. To demonstrate its effectiveness, the detection performance under CMM (9) and SMM (11) are compared, where the parameters of GMM in these two models are consistent. Since there are a large number of parameters to be estimated in model (10), which requires an unacceptable computational cost, the comparison with this model is not in the scope of consideration in this article.

The statistical separability analysis is employed to explore the separation between background and target under two models. The corresponding separation maps are shown in Fig. 4. The blue and red boxes stand for the normalized detection statistics

[1][Online]. Available: http://aviris.jpl.nasa.gov/data/free_data.html
[2][Online]. Available: https://github.com/sxt1996/Cri
[3][Online]. Available: https://github.com/sxt1996/AVIRIS-WTC
[4][Online]. Available: https://github.com/lzw-lzw/Hyperspectral-anomaly-detection-methods/tree/master/DATASET

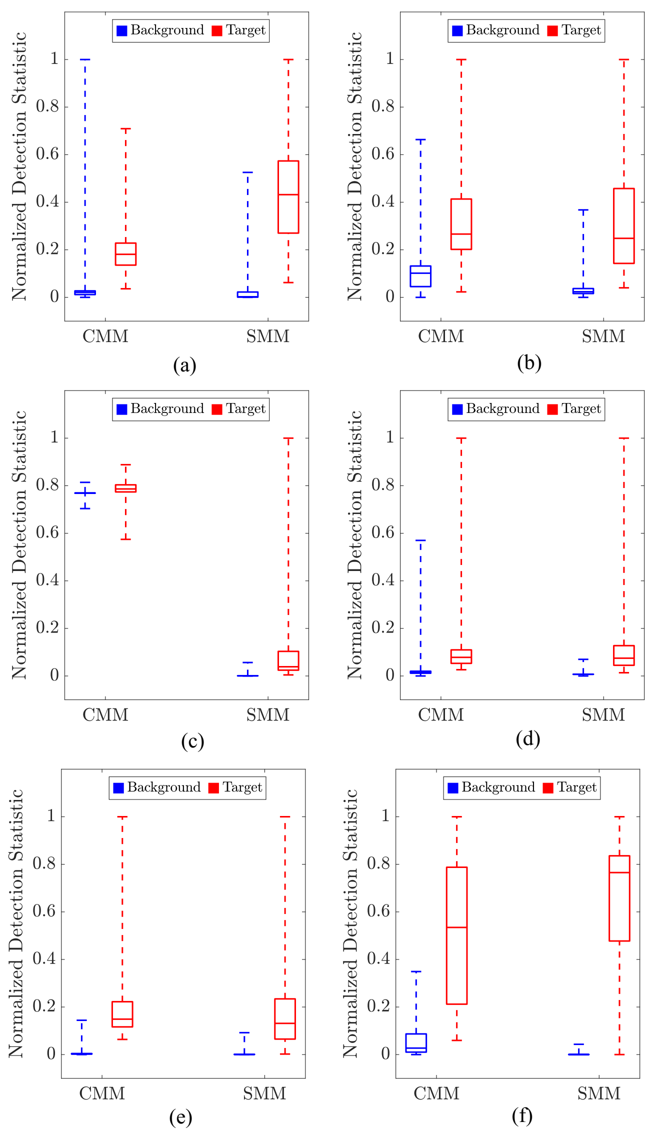


Fig. 4. Separation maps of detections under two models on different HSIs. (a) SDA. (b) Cri. (c) WTC. (d) Hyperion. (e) Pavia. (f) SYN.

of background and target pixels, respectively. The horizontal line in the middle of box represents the mean of the statistic, while the lower and upper boundaries of box indicate the first and third quartiles, respectively.

As illustrated in Fig. 4, the main boxes of background and target under SMM are separated well on all HSIs. The two boxes under CMM in Fig. 4(c) cannot be evidently separated. Meanwhile, since the blue boxes under SMM are compressed quite flat compared with CMM, the SMM is believed to possess better BS. As for Hyperion and Pavia datasets, it seems difficult to distinguish which model has better effect of background–target (BT) separation from Fig. 4(d) and (e). Therefore, both nine AUCs, 2-D and 3-D ROC curves are utilized to further investigate the BT separability and detection performance under two models.

The $\mathrm{AUC}_{\text{TD-BS}}$ reflects the joint effect of TD and BS, which can also characterize BT separability. As shown in the eighth column of Table I, the $\mathrm{AUCs}_{\text{TD-BS}}$ under SMM are superior to CMM except for Cri HSI. Overall, the $\mathrm{AUC}_{(\tau,\text{FPR})}$, AUC_{BS}, and $\mathrm{AUC}_{\text{SNPR}}$ values of SMM are larger than that of CMM without exception. The classical $\mathrm{AUC}_{(\text{TPR, FPR})}$ values under SMM are higher on five HSIs except for Pavia dataset. Especially on the Hyperion, Cri, and WTC HSIs, SMM-based detections have great advantages. Besides, the majority AUC_{TD}, AUC_{OD}, and $\mathrm{AUC}_{\text{ODP}}$ values of SMM are better. In terms of execution time, it can be found in the last column of Table I that the SMM-based GLRT has much less time consumption than CMM-based one on all six experimental images, especially on those spatially large-scale or spectrally high-dimensional HSIs. Meanwhile, the time consumption of both CMM-based and SMM-basd detection is highly correlated with the spatial and spectral sizes of HSI.

As depicted in Fig. 5, the OD performance under SMM is better than that under CMM. In SDA, Cri, Hyperion, and SYN HSIs, adopting the SMM model can achieve significant improvement with respect to detection AUCs. While in WTC and Pavia datasets, there is no obvious contrast between the detection effects under two models. Besides, it can be inferred from the fourth column of Fig. 5 that the SMM-based detector has excellent 2-D ROC curve of (τ, FPR). That is, the FPR of detection quickly drops to 0 when the threshold is low. However, it is difficult to determine which model is better directly from the 2-D ROC curve of (τ, TPR). Further analyzing the $\mathrm{AUC}_{(\tau,\text{TPR})}$ in Table I, the conclusion can be drawn that SMM-based detector outperforms CMM-based one on SDA, Hyperion, and SYN datasets.

In general, it is sufficient to assert that the adopting of SMM instead of CMM can not only significantly improve the detection precision but also effectively enhance BT separation. In addition, the SMM-based subpixel detector possesses the ability to detect targets of any size, shape, and intensity.

C. Detection Performance

In this subsection, the detection performance of the proposed SMGLRT method is compared with classical and new detectors. The detection images of five real and one synthetic HSIs are visually displayed in Fig. 6, where the brightness of pixel in detection images denotes the intensity of target. Intuitively, the proposed SMGLRT detector obtains the best detection results on all HSIs, where the background pixels are dark while the target pixels are bright. Even for large-scale HSIs like Cri, WTC, and SYN, the SMGLRT can still accurately detect subpixel targets. For AMF and CEM detectors, their performance are so close that subtle differences are difficult to visually perceive. In addition, the IRN and CGSAL, corresponding to Fig. 6(e) and (h), cannot separate the target and background clearly. And the TSE fails to accurately locate target pixels on almost all HSIs.

The 2-D ROC curves are shown in Fig. 7 and the corresponding AUCs are presented in Table II. As for AUCs, the SMGLRT acquires the best AUCs in SDA, Cri, Hyperion, Pavia, and SYN datasets. Although the AUC of SMGLRT is 0.022 less than that of AMF and CEM in WTC dataset, it shows stable detection

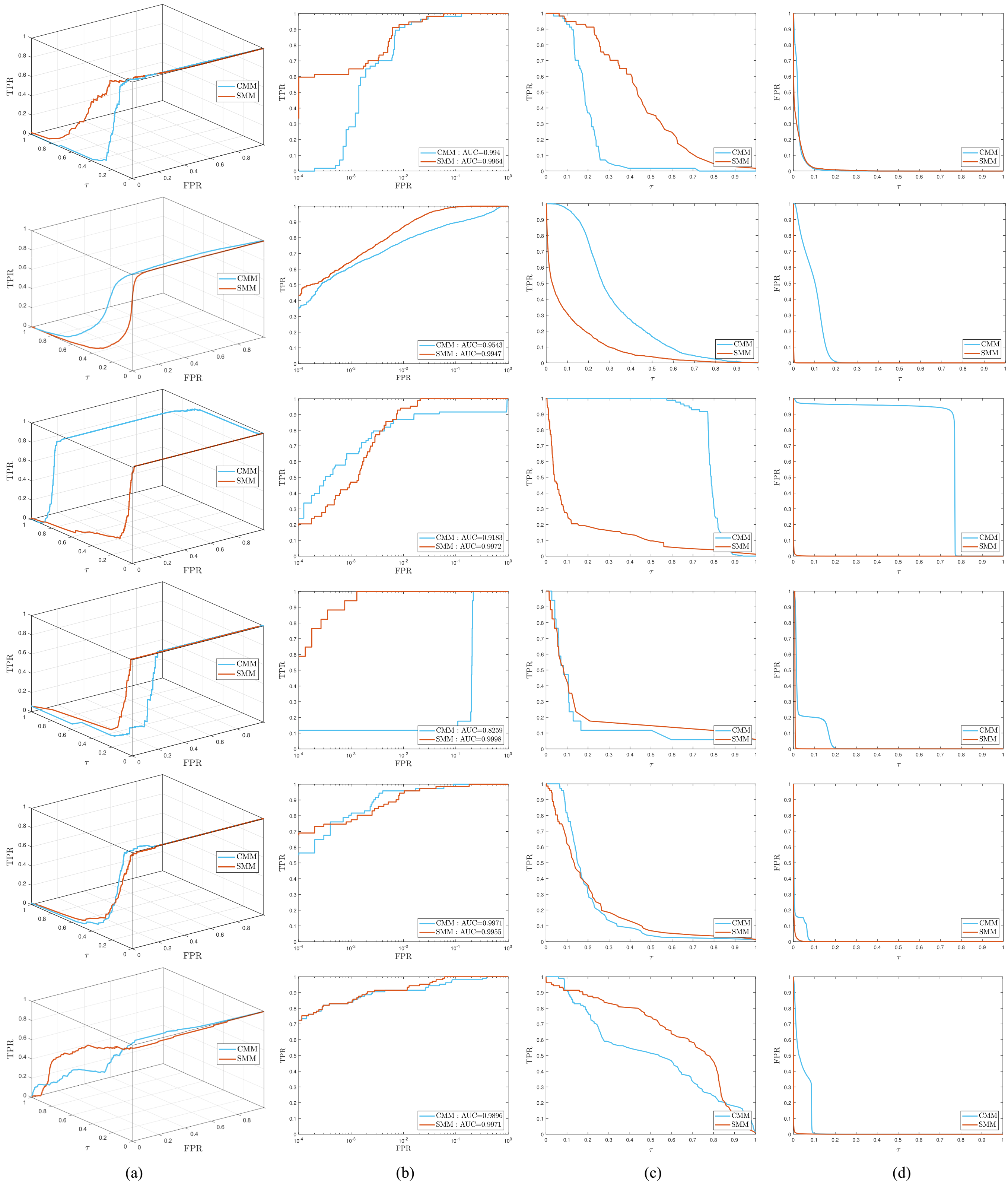
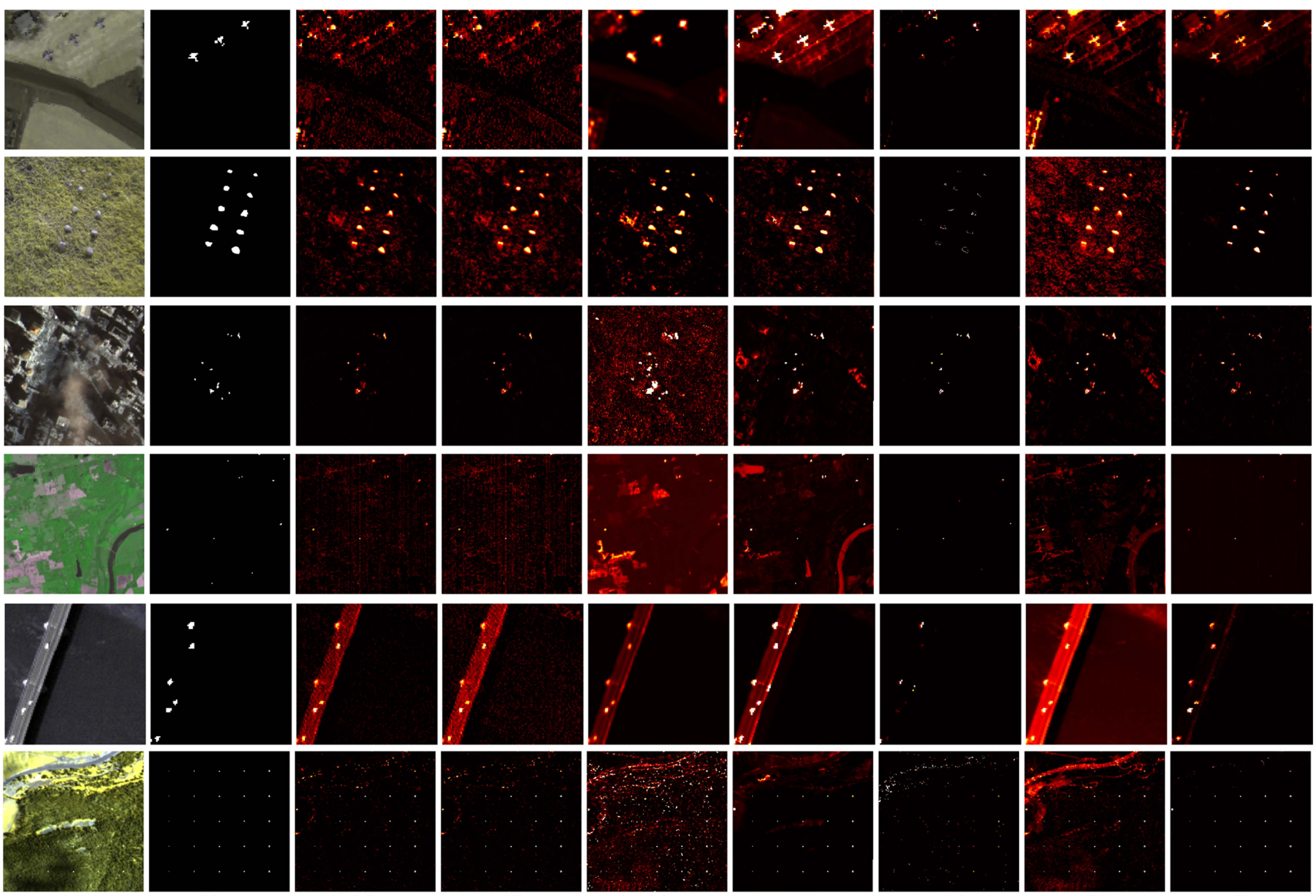


Fig. 5. 3-D and 2-D ROC curves of detectors under CMM (cyan) and SMM (red) on different HSIs, where the 1st to the 6th rows correspond to SDA, Cri, WTC, Hyperion, Pavia adn SYN, respectively. (a) 3-D ROC curves. (b) 2-D ROC curves of (FPR, TPR). (c) 2-D ROC curves of (τ, TPR). (d) 2-D ROC curves of (τ, FPR).

TABLE I
NINE AUCS AND TIME CONSUMPTION UNDER CMM AND SMM

HSIs	Models	AUC									Time(s)
		(TPR, FPR) ↑	(τ, TPR) ↑	(τ, FPR) ↓	TD ↑	BS ↑	TD-BS ↑	ODP ↑	OD ↑	SNPR ↑	
SDA	CMM	0.9940	0.1939	0.0239	1.1879	0.9701	0.1700	1.1700	1.1640	8.1029	1.7739
	SMM	**0.9964**	**0.4472**	**0.0172**	**1.4436**	**0.9792**	**0.4300**	**1.4300**	**1.4264**	**26.0447**	**0.4701**
Cri	CMM	0.9543	**0.3233**	0.0927	**1.2777**	0.8617	**0.2307**	**1.2307**	**1.1850**	3.4887	9.4096
	SMM	**0.9947**	0.1025	**0.0003**	1.0972	**0.9944**	0.1022	1.1022	1.0969	**388.1364**	**2.5313**
WTC	CMM	0.9183	**0.7874**	0.7344	**1.7057**	0.1839	0.0530	1.0530	0.9713	1.0721	55.5049
	SMM	**0.9972**	0.1404	**0.0011**	1.1375	**0.9961**	**0.1393**	**1.1393**	**1.1365**	**130.5749**	**8.8211**
Hyperion	CMM	0.8259	0.1627	0.0450	0.9886	0.7809	0.1177	1.1177	0.9436	3.6161	4.2704
	SMM	**0.9998**	**0.2036**	**0.0077**	**1.2033**	**0.9921**	**0.1959**	**1.1959**	**1.1956**	**26.4502**	**1.8162**
Pavia	CMM	**0.9971**	**0.1998**	0.0124	**1.1968**	0.9847	0.1874	1.1874	1.1845	16.1432	0.3290
	SMM	0.9955	0.1991	**0.0020**	1.1946	**0.9935**	**0.1971**	**1.1971**	**1.1926**	**100.5973**	**0.2730**
SYN	CMM	0.9896	0.5083	0.0434	1.4979	0.9462	0.4649	1.4649	1.4545	11.7016	116.7433
	SMM	**0.9971**	**0.6422**	**0.0008**	**1.6393**	**0.9963**	**0.6414**	**1.6414**	**1.6385**	**793.7998**	**12.7654**

The ↑ and ↓ symbols denote the larger the better and the smaller the better, respectively. The bold entities represent the best results.



(a) (b) (c) (d) (e) (f) (g) (h) (i)

Fig. 6. Detection images of different detectors. (a) False color images of SDA (the 1st row), Cri (the 2nd row), WTC (the 3 rd row), Hyperion (the 4th row), Pavia (the 5th row), and Synthetic scene (the 6th row). (b) Groundtruth maps. (c) AMF. (d) CEM. (e) IRN. (f) TSTTD. (g) TSE. (h) CGSAL. (i) SMGLRT.

performance according to its detection image in the third row of Fig. 6(i). It is worth noting that the performance of AMF and CEM detectors is not stable enough, even though they obtain terrific detection AUCs on WTC datasets.

From Fig. 7(a) and (e), we can see that the proposed SMGLRT shows excellent performance than other detectors. However, for scenes with complex background (Fig. 7(c) and (d)), the SMGLRT seems weak to accurately detect the target at a low

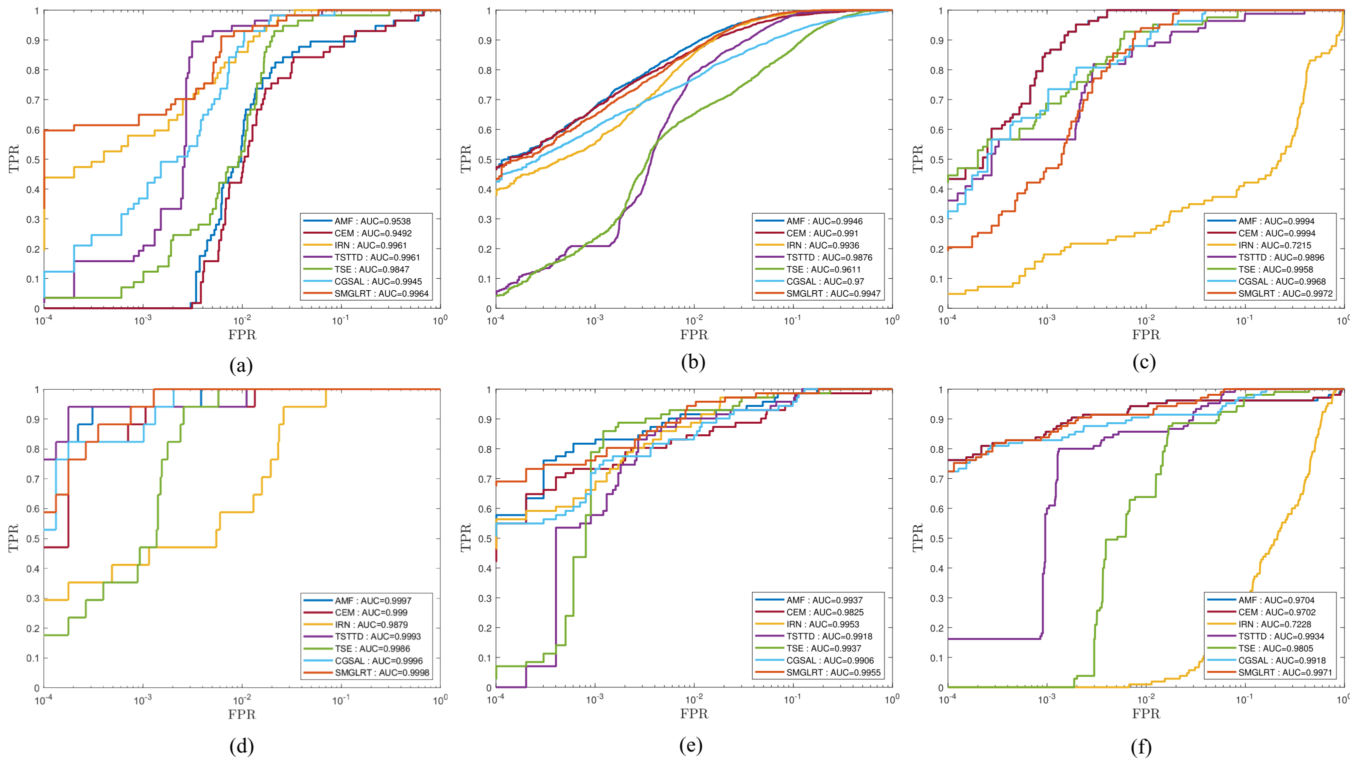


Fig. 7. 2-D ROC curves of detectors on different HSIs. (a) SDA. (b) Cri. (c) WTC. (d) Hyperion. (e) Pavia. (f) SYN.

TABLE II
DETECTION AUCs OF DIFFERENT DETECTORS

HSIs (Size)	Methods						
	AMF	CEM	IRN	TSTTD	TSE	CGSAL	SMGLRT
SDA (100×100)	0.9538	0.9492	0.9961	0.9961	0.9847	0.9945	**0.9964**
Cri (400×400)	0.9946	0.9910	0.9936	0.9876	0.9611	0.9700	**0.9947**
WTC (200×200)	**0.9994**	**0.9994**	0.7215	0.9896	0.9958	0.9968	0.9972
Hyperion (150×150)	0.9997	0.9990	0.9879	0.9993	0.9986	0.9996	**0.9998**
Pavia (100×100)	0.9937	0.9825	0.9953	0.9918	0.9937	0.9906	**0.9955**
SYN (280×280)	0.9704	0.9702	0.7228	0.9934	0.9805	0.9918	**0.9971**

The bold entities represent the best results.

FPR. Except for these two HSIs, the SMGLRT gets a pretty high TPR when the FPR is at the level of 10^{-4}.

V. CONCLUSION

In this work, an SMGLRT detector has been developed on the basis of SMM. Distinguishing from CMM, the SMM can accurately depict the mixing extent on different bands while restraining the model complexity. Under the assumption of Gaussian-mixture-distributed background, the SMM is established for the task of hyperspectral target detection, and the optimal segmentation is acquired from the perspective of block-diagonalization for covariance matrix. Moreover, the segmented mixing coefficients are estimated and then the GLRT statistic is derived. The experiments on real and synthetic HSIs demonstrate the superiority of SMGLRT over the compared detectors, and also confirm that replacing CMM with SMM significantly enhance detection stability.

The vital contribution of this work is providing an insight into mixing pattern between background and target. Assigning different mixing coefficients to distinct segments is to strive for a balance between the generalization and sophistication of model. A moderate model can not only improve detection performance but also adapt to different scenarios. Therefore, the idea of segmented mixing can be also extended to hyperspectral unmixing task. Besides, it is a worthwhile extension to consider the impact of different spectral clusters on detection performance.

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